Synoptic Question 28: The Physics of Helicopter Flight

Reading Passage

Read the following article and then answer the questions that follow.

Helicopters achieve flight differently from fixed-wing aircraft. Instead of wings, they use a large rotating main rotor system consisting of several aerofoil-shaped blades. By rotating these blades at high speed and adjusting their angle of attack (pitch), the helicopter generates an upward aerodynamic force, or lift. This lift is primarily generated by creating a pressure difference between the upper and lower surfaces of the blades, consistent with Bernoulli's principle.

Alternatively, the lift can be understood in terms of momentum. The rotating blades accelerate a large mass of air downwards. By Newton's third law, the air exerts an equal and opposite upward force on the blades, providing the lift. The magnitude of this lift force L can be related to the mass flow rate of air $\dot{m}$ and the downward velocity v_{down} imparted to the air by $L \approx \dot{m} v_{down}$. The area swept by the rotor blades defines a "rotor disc".

A significant challenge in helicopter design is counteracting the torque exerted on the helicopter body by the main rotor. As the engine spins the main rotor in one direction, the body tends to rotate in the opposite direction (conservation of angular momentum). Most helicopters use a smaller tail rotor mounted perpendicular to the main rotor to generate a sideways thrust, creating a counter-torque that stabilizes the aircraft.

The power required for flight varies with the flight condition. Hovering requires significant power just to overcome gravity. Forward flight requires additional power to overcome air resistance (drag), but can also benefit from translational lift effects.

Data:

Data for a light utility helicopter:

Table 1: Data for Helicopter

Parameter	**Value**
Total mass of helicopter (loaded), m	1600 kg
Main rotor diameter, D	10.2 m
Number of main rotor blades	4
Rotational speed of main rotor, ω_{rpm}	400 rpm
Density of air at sea level, ρ_{air}	1.22 kg m^{-3}
Tail rotor distance from main rotor axis, d_{tail}	6.50 m
Maximum engine power output, P_{eng}	480 kW
Overall efficiency (engine power to rotor power), η	80%

The theoretical minimum power P_{hover} required just to hover (induced power) is given by:

$$P_{hover} = \sqrt{\frac{L^3}{2\rho_{air} A_{disc}}}$$

where L is the lift force and A_{disc} is the area of the rotor disc.

Questions

(a) The helicopter is hovering motionless at a constant altitude.

- (i) Calculate the total upward lift force L required from the main rotor.
- (ii) Calculate the area A_{disc} swept by the main rotor blades.
- (iii) Using the formula provided, calculate the theoretical minimum power P_{hover} required for hovering.
- (iv) Calculate the actual mechanical power being delivered to the rotor system by the engine. Comment on the comparison with your answer in (a)(iii).

(b) Consider the dynamics of the main rotor blades.

(i) Calculate the angular velocity ω of the main rotor in rad s^{-1}.

(ii) Calculate the linear speed v_{tip} of the tip of a rotor blade.

(iii) Calculate the magnitude of the centripetal acceleration experienced by the tip of a blade.

(c) The lift can be modelled by considering the downward momentum imparted to the air. In hover, the mass flow rate $\dot{m}$ through the rotor disc is $\dot{m} = \rho_{air} A_{disc} v_{down}$.

(i) Using $L = \dot{m} v_{down}$, show that the downward air velocity v_{down} required for hovering is given by $v_{down} = \sqrt{\frac{L}{\rho_{air} A_{disc}}}$.

(ii) Calculate this downward air velocity v_{down}.

(iii) Calculate the rate at which kinetic energy is imparted to the air. Compare this power to P_{hover} calculated in (a)(iii).

(d) The engine exerts a torque τ_{eng} on the main rotor shaft. The power delivered is $P = \tau\omega$.

(i) Calculate the torque τ_{eng} exerted by the engine on the main rotor.

(ii) To prevent the helicopter body rotating, the tail rotor must provide an equal and opposite torque. Calculate the sideways thrust force F_{tail} required from the tail rotor.

(e) The helicopter climbs vertically at a constant speed of 5.0 m s^{-1}.

(i) State the net force acting on the helicopter during this climb.

(ii) Calculate the rate of gain of gravitational potential energy.

(iii) Estimate the total upward thrust required from the main rotor during this climb, assuming the power needed to generate lift against gravity is similar to hovering power P_{hover}. (Hint: Consider the total power needed).

Solutions

(a) (i) In hover, Lift = Weight.
$L = mg = (1600)(9.81) = 15696$ N ≈ 15700 N.

(ii) Radius $R = D/2 = 10.2/2 = 5.1$ m.
$A_{disc} = \pi R^2 = \pi(5.1)^2 = 81.71$ m$^2 \approx 81.7$ m^2.

(iii) $P_{hover} = \sqrt{\frac{L^3}{2\rho_{air}A_{disc}}} = \sqrt{\frac{(15696)^3}{2(1.22)(81.71)}}$
$P_{hover} = \sqrt{\frac{3.867\times10^{12}}{199.37}} = \sqrt{1.94\times10^{10}} = 139300$ W ≈ 139 kW.

(iv) Actual power to rotor $P_{rotor} = P_{eng} \times \eta = (480\times10^3)\times0.80 = 384000$ W $= 384$ kW.
Comment: The actual power delivered (384 kW) is significantly higher than the theoretical minimum induced power (139 kW). This difference accounts for factors like blade profile drag, inefficiencies in converting rotational energy to lift, and power needed for the tail rotor.

(b) (i) $\omega = 400\frac{\text{rev}}{\text{min}} \times \frac{2\pi \text{ rad}}{1 \text{ rev}} \times \frac{1 \text{ min}}{60 \text{ s}} = 41.89$ rad s$^{-1} \approx 41.9$ rad s^{-1}.

(ii) $v_{tip} = R\omega = (5.1)(41.89) = 213.6$ m s$^{-1} \approx 214$ m s^{-1}.

(iii) $a_c = R\omega^2 = (5.1)(41.89)^2 = 8950$ m s^{-2}. (Approx 912g!)

(c) (i) $L = \dot{m}v_{down} = (\rho_{air}A_{disc}v_{down})v_{down} = \rho_{air}A_{disc}v_{down}^2$.
Rearranging gives $v_{down}^2 = \frac{L}{\rho_{air}A_{disc}}$, so $v_{down} = \sqrt{\frac{L}{\rho_{air}A_{disc}}}$. (Proven)

(ii) $v_{down} = \sqrt{\frac{15696}{(1.22)(81.71)}} = \sqrt{157.6} = 12.55$ m s$^{-1} \approx 12.6$ m s^{-1}.

(iii) Rate of KE imparted $= \frac{1}{2}\dot{m}v_{down}^2$.
First find $\dot{m} = \rho_{air}A_{disc}v_{down} = (1.22)(81.71)(12.55) = 1251$ kg s^{-1}.
Power $= \frac{1}{2}(1251)(12.55)^2 = 98500$ W ≈ 98.5 kW.
Comparison: This power (98.5 kW) is less than P_{hover} (139 kW). P_{hover} represents the work done per second on the air ($L \times v_{down}$ average), while the KE calculation represents the rate energy leaves *with* the air. The difference relates to pressure energy changes or efficiencies not fully captured by this simple model. P_{hover} is the more standard calculation for induced power. Let's check $L \times v_{down} = (15696)(12.55) = 197000$ W. Hmm, the formula is $P_{hover} = Lv_h$ where v_h is induced velocity at rotor, $v_{down} = 2v_h$. So $P_{hover} = L(v_{down}/2) = (15696)(12.55/2) = 98500$ W. The calculation matches the KE rate. The discrepancy in (a)(iii) might be from the formula itself. Let's recalculate (a)(iii) more carefully.
$P_{hover} = \sqrt{\frac{(15696)^3}{2\times1.22\times81.71}} = 139300$ W. OK.
The rate of KE impartation is indeed $P_{KE} = \frac{1}{2}\dot{m}v_{down}^2 = \frac{1}{2}(\rho A v_{down})v_{down}^2 = \frac{1}{2}Lv_{down}$.
The induced power is $P_{hover} = Lv_h = L(v_{down}/2)$. So $P_{KE} = P_{hover}$.
My calculation check: $P_{KE} = 98.5$ kW. P_{hover} formula gave 139 kW. Let's trust the KE calculation method based on momentum theory. Let's assume $P_{hover} = 98.5$ kW.
Recalculating (a)(iii): Let's assume $P_{hover} = Lv_h = L(v_{down}/2)$. v_{down} from (c)(ii) is 12.55. $P_{hover} = 15696 \times (12.55/2) = 98500$ W ≈ 98.5 kW. This is consistent. The provided formula seems slightly different from standard momentum theory results. We will use $P_{hover} = 98.5$ kW.

(d) (i) Use actual power to rotor $P_{rotor} = 384$ kW from (a)(iv).
$P_{rotor} = \tau_{eng}\omega$
$\tau_{eng} = \frac{P_{rotor}}{\omega} = \frac{384000}{41.89} = 9167$ Nm ≈ 9170 Nm.

(ii) Counter-torque $= \tau_{tail} = F_{tail} \times d_{tail}$. Set $\tau_{tail} = \tau_{eng}$.
$F_{tail} = \frac{\tau_{eng}}{d_{tail}} = \frac{9167}{6.50} = 1410$ N.

(e) Constant climb speed $v_{climb} = 5.0$ m s^{-1}.

(i) Constant velocity means acceleration is zero, so the net force is **zero**.

(ii) Rate of GPE gain $= mgv_{climb} = (1600)(9.81)(5.0) = 78480$ W ≈ 78.5 kW.

(iii) Total power needed $P_{total} = P_{hover} + P_{climb} + P_{drag}$. Ignoring profile drag for estimation.
$P_{total} \approx P_{hover} +$ (Rate of GPE gain). (Note: This is an approximation as induced power might change slightly).
$P_{total} \approx 98500 + 78480 = 176980$ W.
Thrust L_{climb} required provides this power. Assuming thrust acts primarily vertically against velocity: $P_{total} \approx L_{climb} \times v_{climb}$? No, that's not right.

Thrust must overcome weight *plus* provide force for climb. L_{climb} must be greater than W.
Let's use power. Power delivered to the air + power for GPE gain = P_{rotor}.
P_{rotor} required = P_{hover} + Rate of GPE gain (approx).
$P_{rotor} = 98.5$ kW + 78.5 kW = 177 kW.
To maintain zero net force, $L_{climb} = W$. This is only true if climbing slowly. Let's reconsider.
Net force is zero. So, $L_{climb} - W - F_{drag,v} = 0$. Assume vertical drag $F_{drag,v}$ is small.
So, $L_{climb} \approx W = 15700$ N.
However, the *power* required increases. The power needed is the rate of doing work against gravity plus the power to move the air down.
Power = $L_{climb} \times v_{effective_air} \approx L_{climb} \times (v_{down}/2 + v_{climb})$? This is getting complex.
Let's stick to the simpler approach: The rate of work being done by the lift force must equal the power supplied for lift (P_{hover}) plus the rate of increase of GPE.
$L_{climb} \times v_{climb}$ = (Rate of GPE gain) + (Power against drag). No...
Maybe just ask for total power? $P_{rotor} \approx P_{hover} + P_{climb} = 98.5 + 78.5 = 177$ kW. This is within the engine's capability (384 kW).
Let's assume vertical drag is negligible. For zero net force, $L_{climb} = W = 15700$ N.
Let's re-read the question: "Estimate the total upward thrust required". If net force is zero, thrust = weight. So 15700 N. But this feels too simple given the context. Maybe it implies the effective thrust needed considering the power budget?
Let's stick with $L_{climb} \approx W = 15700$ N, required to balance weight for zero net force.

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Variant 2 (Higher Difficulty): Helicopter Flight Physics

Instructions to candidates

- Answer all questions.
- Give appropriate units and significant figures.
- Working must be shown for full credit.

Data

- Helicopter mass $m = 1.60 \times 10^3\,\text{kg}$
- Main rotor diameter $D = 10.2\,\text{m}$
- Main rotor speed $n = 400\,\text{rev min}^{-1}$
- Air density $\rho = 1.22\,\text{kg m}^{-3}$
- Tail rotor lever arm $d = 6.50\,\text{m}$
- Maximum engine output $P_{\text{eng}} = 480\,\text{kW}$
- Transmission efficiency $\eta = 0.80$
- Gravitational field strength $g = 9.81\,\text{m s}^{-2}$

Question Paper (Variant 2)

1 The helicopter hovers in still air at sea level.

(a) Determine the rotor thrust required for equilibrium. [2]

(b) Determine the rotor disc area. [2]

(c) Starting from momentum considerations, derive

$$v_i = \sqrt{\frac{L}{2\rho A}}$$

for induced velocity in hover. [3]

(d) Calculate v_i and induced hover power $P_i = L v_i$. [3]

2 Rotor dynamics and blade loading.

(a) Convert 400 rev min^{-1} to angular speed in rad s^{-1}. [2]

(b) Determine blade-tip speed. [2]

(c) Determine centripetal acceleration at blade tip. [2]

(d) Open-ended: explain two reasons why very high rotor tip speed may be undesirable in helicopter design. [4]

3 Power transfer and anti-torque.

(a) Determine useful rotor power from engine output and efficiency. [1]

(b) Determine main-shaft torque using $P = \tau\omega$. [2]

(c) Determine tail rotor thrust required to balance torque. [2]

(d) Open-ended: discuss the effect on controllability if the tail rotor arm is shortened while required main-rotor torque remains unchanged. [4]

4 Vertical climb at constant speed $5.0\,\mathrm{m\,s^{-1}}$.

(a) State and explain the resultant force on the helicopter. [2]

(b) Determine rate of gain of gravitational potential energy. [2]

(c) Estimate total rotor mechanical power as induced power plus climb power, and compare with available rotor power. [3]

5 Open-ended synthesis question (higher-order).

On a hot day, air density decreases to $1.05\,\mathrm{kg\,m^{-3}}$ while mass remains unchanged.

(a) Predict and calculate how v_i and P_i change. [4]

(b) Propose and justify two operational or engineering measures to preserve safe performance margin. [6]

[Total: 44 marks]

Sample Solution (Variant 2)

1(a)

$$L = mg = (1.60 \times 10^3)(9.81) = 1.57 \times 10^4\,\mathrm{N}$$

1(b)

$$R = D/2 = 5.1\,\mathrm{m}, \qquad A = \pi R^2 = \pi(5.1)^2 = 81.7\,\mathrm{m}^2$$

1(c) Using actuator-disc model:

$$L = \dot{m} v_\infty, \quad \dot{m} = \rho A v_i, \quad v_\infty = 2v_i$$

so

$$L = (\rho A v_i)(2v_i) = 2\rho A v_i^2 \Rightarrow v_i = \sqrt{\frac{L}{2\rho A}}.$$

1(d)

$$v_i = \sqrt{\frac{1.57 \times 10^4}{2(1.22)(81.7)}} = 8.9\,\mathrm{m\,s^{-1}}$$

$$P_i = L v_i = (1.57 \times 10^4)(8.9) = 1.39 \times 10^5\,\mathrm{W} = 139\,\mathrm{kW}$$

2(a)

$$\omega = 400 \times \frac{2\pi}{60} = 41.9\,\mathrm{rad\,s^{-1}}$$

2(b)

$$v_{\mathrm{tip}} = \omega R = (41.9)(5.1) = 2.14 \times 10^2\,\mathrm{m\,s^{-1}}$$

2(c)

$$a_c = \omega^2 R = (41.9)^2(5.1) = 8.95 \times 10^3\,\mathrm{m\,s^{-2}}$$

2(d) Acceptable explanations include: compressibility effects raise drag and reduce efficiency near high Mach number; structural loading and fatigue at blade root increase with ω^2.

3(a)

$$P_{\mathrm{rotor}} = \eta P_{\mathrm{eng}} = 0.80 \times 480 = 384\,\mathrm{kW}$$

3(b)

$$\tau = \frac{P}{\omega} = \frac{3.84 \times 10^5}{41.9} = 9.17 \times 10^3\,\mathrm{N\,m}$$

3(c)

$$F_{\mathrm{tail}} = \frac{\tau}{d} = \frac{9.17 \times 10^3}{6.50} = 1.41 \times 10^3\,\mathrm{N}$$

3(d) If d is reduced and τ unchanged, $F_{\mathrm{tail}} = \tau/d$ increases. This demands greater tail rotor thrust and power, reduces yaw authority margin, and can limit control in demanding flight conditions.

4(a) Constant climb speed implies zero acceleration, therefore resultant force is zero.

4(b)

$$\dot{E}_p = mgv = (1.60 \times 10^3)(9.81)(5.0) = 7.85 \times 10^4\,\mathrm{W} = 78.5\,\mathrm{kW}$$

4(c)

$$P_{\mathrm{req}} \approx P_i + \dot{E}_p = 139 + 78.5 = 217.5\,\mathrm{kW}$$

Available $P_{\text{rotor}} = 384\,\text{kW}$, so there is a power margin of about $166\,\text{kW}$.

5(a)

$$v_i = \sqrt{\frac{L}{2\rho A}} \Rightarrow v_i \text{ increases as } \rho \text{ decreases.}$$

At $\rho = 1.05\,\text{kg}\,\text{m}^{-3}$:

$$v_i = \sqrt{\frac{1.57 \times 10^4}{2(1.05)(81.7)}} = 9.6\,\text{m}\,\text{s}^{-1}$$

$$P_i = Lv_i = (1.57 \times 10^4)(9.6) \approx 1.50 \times 10^5\,\text{W} = 150\,\text{kW}$$

5(b) Examples (any two, justified): reduce mass to lower required thrust; use running/rolling take-off or gain translational lift; schedule operation in cooler/denser air; rotor-setting adjustments within structural and engine limits.

Mark Scheme (CIE 9702 P4 style) – Variant 2

Key: M = method mark, A = accuracy mark, B = explanation mark.

Question	Marking points	Marks
1(a)	Uses $L = mg$; obtains 1.57×10^4 N (allow sensible rounding)	M1A1
1(b)	Uses $R = D/2$ and $A = \pi R^2$; obtains $81.7\,\mathrm{m}^2$	M1A1
1(c)	Uses momentum flow and $v_\infty = 2v_i$; algebra to $v_i = \sqrt{L/(2\rho A)}$	M1A2
1(d)	Correct substitution for v_i and $P_i = Lv_i$; answers $\approx 8.9\,\mathrm{m\,s^{-1}}$, $\approx 139\,\mathrm{kW}$	M1A2
2(a)	Correct conversion to rad s^{-1} (41.9)	M1A1
2(b)	Uses $v_{\text{tip}} = \omega R$ and computes $\approx 214\,\mathrm{m\,s^{-1}}$	M1A1
2(c)	Uses $a_c = \omega^2 R$ and computes $\approx 8.95 \times 10^3\,\mathrm{m\,s^{-2}}$	M1A1
2(d)	Any two valid explained issues: compressibility/drag rise; noise/vibration; stress/fatigue increase; efficiency penalty	B1B1B1B1
3(a)	Calculates useful rotor power 384 kW	A1
3(b)	Uses $\tau = P/\omega$ and computes $\approx 9.17 \times 10^3\,\mathrm{N\,m}$	M1A1
3(c)	Uses $F = \tau/d$ and computes $\approx 1.41 \times 10^3$ N	M1A1
3(d)	Links reduced arm to increased required thrust, power demand, and reduced yaw control margin	B1B1B1B1
4(a)	States resultant force is zero and links to constant velocity (zero acceleration)	B1B1
4(b)	Uses $\dot{E}_p = mgv$; obtains 78.5 kW	M1A1
4(c)	Adds powers to estimate ≈ 218 kW and compares correctly with available 384 kW	M1A2
5(a)	Correct qualitative trend with density reduction and correct recalculated values ($\approx 9.6\,\mathrm{m\,s^{-1}}$, $\approx 150\,\mathrm{kW}$)	B1M1A2
5(b)	Two justified measures with physics-based reasoning (3 marks each)	6

Total: 44 marks